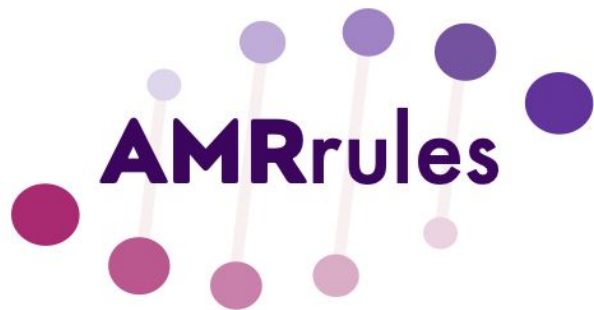


ESGEM-AMR



ESCMID



amrrules.org

Agenda

1. Updates from subgroups
2. Rules with no experimental/statistical evidence
3. Code updates
4. Manuscript planning

Updates from subgroups

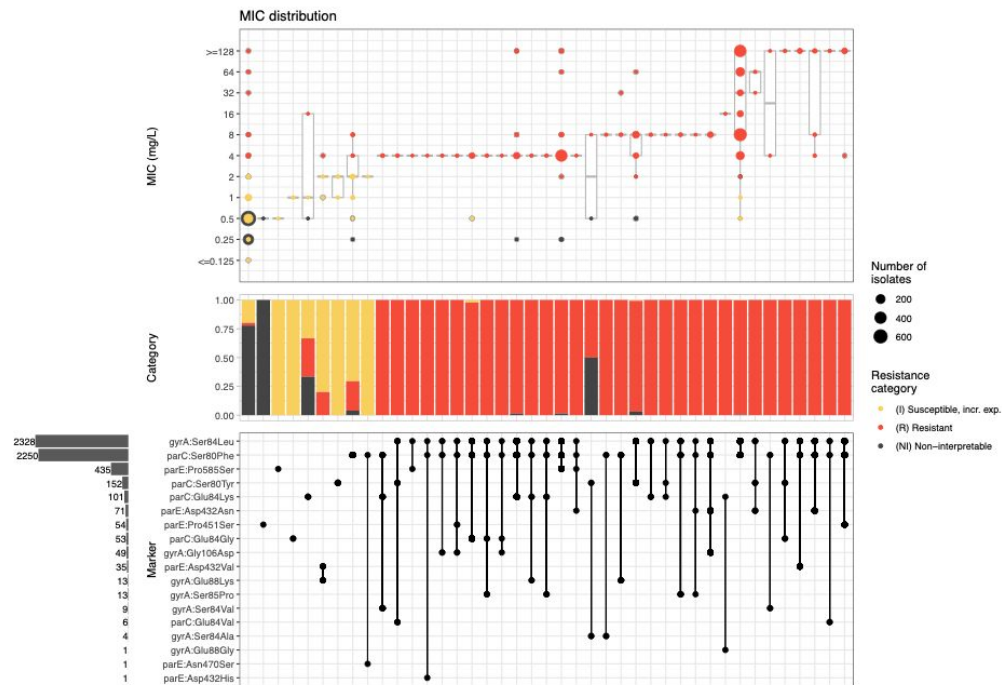
- *Staphylococcus* (Natacha Couto)
- *Acinetobacter* (Paul Higgins)
- *Legionella* (Ghislaine Descours)
- *Enterococcus* (Francesc Coll)
- *Burkholderia pseudomallei* (Claire Chewapreecha)
- *Neisseria gonorrhoeae* (Leonor Sanchez-Buso)
- *Achromobacter xylosoxidans* (Charlotte Michel)

***Staphylococcus* subgroup**

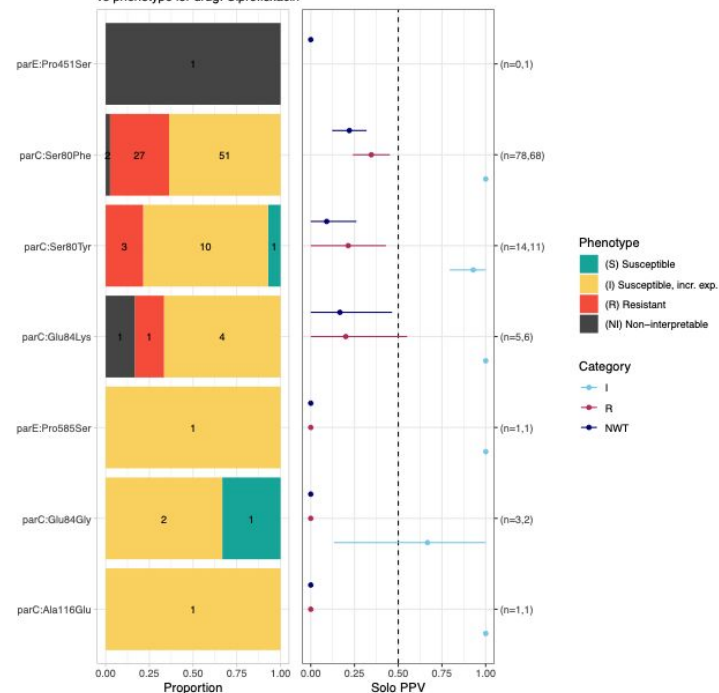
Natacha Couto, Sarah Baines, Sandra Reuter, Birgit Strommenger, Birgitta Duim, Tim Read, Lina Cavaco, Valeria Bortolaia, Sabrina di Gregorio, Robert Kozak, Raphael Sieber, Teresa Ribeiro, Sankarganesh Jeyaraj, Manal AbuOun, Holly Grace Espiritu

Staphylococcus subgroup

- We initially created 64 rules for expected resistances or core genes in *S. aureus*.
- Gathered 9299 genotype-phenotype observations (54 antimicrobials) from multiple studies and countries.



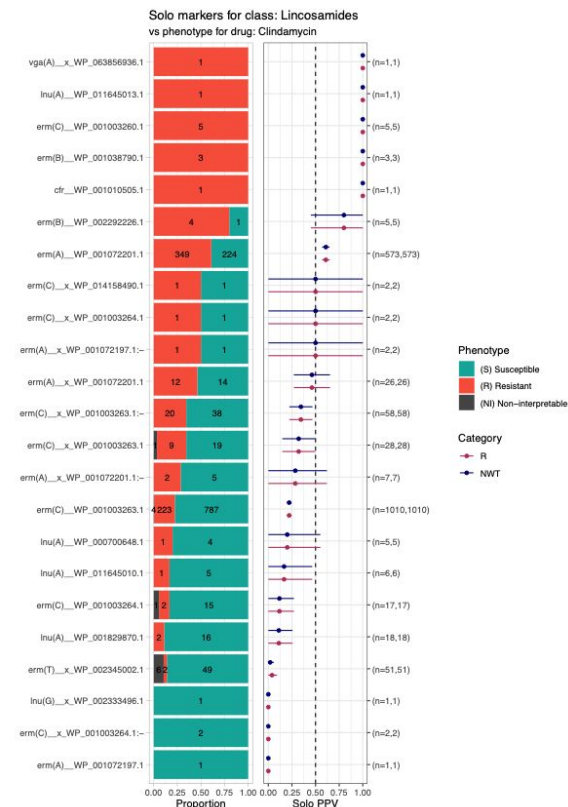
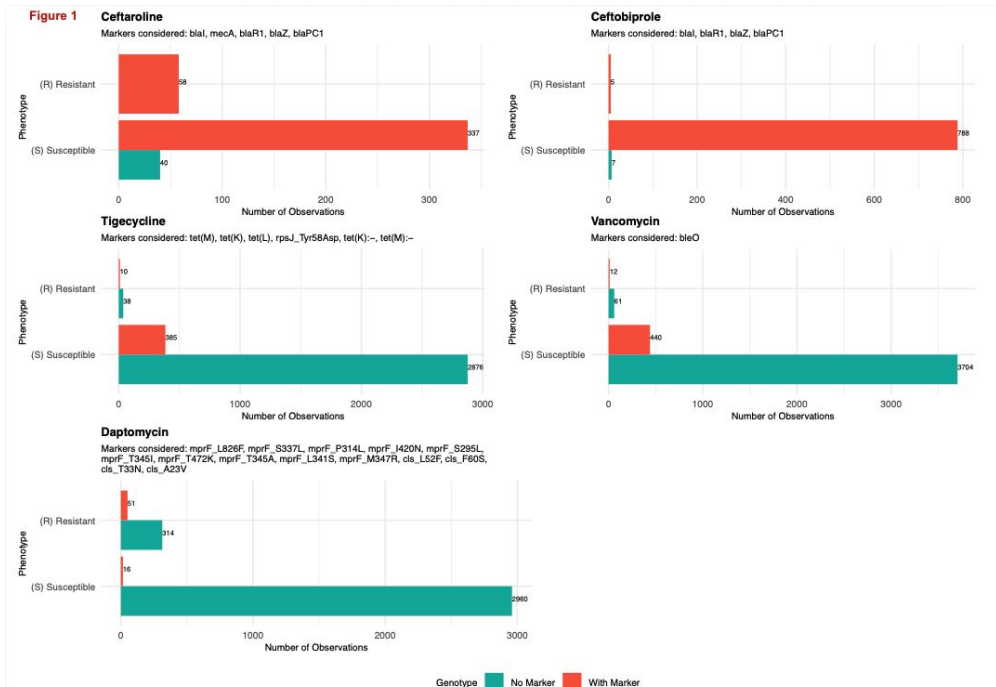
**Solo markers for class: Quinolones
vs phenotype for drug: Ciprofloxacin**



Staphylococcus subgroup

- After running AMRrulemaker, we have generated 300 new rules for 20 antimicrobials.
- Curation is now ongoing with weekly meetings.

Staphylococcus subgroup



Acinetobacter subgroup

Members:

- Bogdan Iorga
- Mehrad Hamidian
- Bruno Silvester Lopes
- Margaret Lam
- Balint Kintses
- Ignasi Roca
- Valeria Bortolaia
- Paul Higgins
- Antoine Abou Fayad
- Rahul Garg
- Priyanka Khopkar-Kale
- Varun Shamanna
- Clement Tsui
- David Wareham

Activities to date:

- Monthly online meetings
- Google account and Google Drive for file sharing
- Google sheets for adding/editing resistance genes/susceptibility data
- Discuss the different sequence data output i.e. long vs short
- Nature of intrinsic vs acquired resistance
- List of core genes from *A. baumannii*
 - What variants are important?
 - Role of mutations and IS elements and how to incorporate these data
- Greater emphasis on using resistance mechanisms determined from clinical isolates opposed to lab-grown mutants
- All rules are approved by at least three members of the group

ESGEM-AMR Working Group: *Acinetobacter*

Members:

- Bogdan Iorga
- Mehrad Hamidian
- Bruno Silvester Lopes
- Margaret Lam
- Balint Kintses
- Ignasi Roca
- Valeria Bortolaia
- Paul Higgins
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Class	Examples	Note	Mechanism
Penicillin	Ampicillin, amoxicillin	Intrinsic resistance	ADC
Cephalosporins	Ceftazidime	Intrinsic resistance	ADC
Carbapenems	Imipenem, meropenem	Acquired resistance	OXA, NDM
Monobactams	Aztreonam	Intrinsic resistance	
Fluoroquinolones	Ciprofloxacin, levofloxacin	Acquired resistance	<i>gyrA/parC</i> mutations
Aminoglycosides	Amikacin, gentamicin, tobramycin	Acquired resistance	Modifying enzymes
Glycopeptides		Intrinsic resistance	
Macrolides		Intrinsic resistance	Various efflux pumps
Tetracyclines		Acquired resistance	
Oxazolidinones		Intrinsic resistance	
Miscellaneous agents	Colistin/polymyxin B	Acquired resistance	PmrCAB
	Trimethoprim-sulfamethoxazole		Sul, efflux
	Tigecycline		Efflux, <i>tet</i> -variants
	Cefiderocol		PirA/PBP3, PER

Red = breakpoints set

Green = no breakpoints but there is interest in these drugs

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Intrinsic OXA

A. baumannii resistance genes							
File Edit View Insert Format Data Tools Extensions Help							
Menus							
A1							
organism							
	A	B	C	D	E	F	G
1	organism	gene	context	drug	category	PMID	rule curation note
2	A. baumannii	blaOXA-51 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15649299	Narrow spectrum carbapenem resistance - confers resistance to carbapenems if the expression enhanced by the insertion of an insertion sequence (often ISAb1, ISAb125 etc)
3	A. baumannii	blaOXA-64 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	
4	A. baumannii	blaOXA-65 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	
5	A. baumannii	blaOXA-66 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	Narrow spectrum carbapenem resistance - confers resistance to carbapenems if the expression enhanced by the insertion of an insertion sequence (often ISAb1, ISAb125 etc)
6	A. baumannii	blaOXA-67 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
7	A. baumannii	blaOXA-68 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	
8	A. baumannii	blaOXA-69 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	Narrow spectrum carbapenem resistance - confers resistance to carbapenems if the expression enhanced by the insertion of an insertion sequence (often ISAb1, ISAb125 etc)
9	A. baumannii	blaOXA-70 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	
10	A. baumannii	blaOXA-71 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	15760431	
11	A. baumannii	blaOXA-75 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	16189095	
12	A. baumannii	blaOXA-76 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
13	A. baumannii	blaOXA-77 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
14	A. baumannii	blaOXA-78 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
15	A. baumannii	blaOXA-79 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17850339	
16	A. baumannii	blaOXA-80 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17850339	
17	A. baumannii	blaOXA-82 (blaOXA-51/oxaAb family)	core	Carbapenems	nwt R	33338207, 1785033	
18	A. baumannii	blaOXA-83 (blaOXA-51/oxaAb family)	core	Carbapenems	wt I	33338207, 1663025	
19	A. baumannii	blaOXA-84 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	16630258, 1761060	
20	A. baumannii	blaOXA-86 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	16816400	
21	A. baumannii	blaOXA-87 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	16816400	
22	A. baumannii	blaOXA-88 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17284537	
23	A. baumannii	blaOXA-89 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	16844350	
24	A. baumannii	blaOXA-90 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	19567349	
25	A. baumannii	blaOXA-91 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17284537	
26	A. baumannii	blaOXA-92 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17391399	
27	A. baumannii	blaOXA-93 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17284537	
28	A. baumannii	blaOXA-94 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17284537	
29	A. baumannii	blaOXA-95 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17284537	
30	A. baumannii	blaOXA-98 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
31	A. baumannii	blaOXA-99 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	19394202	
32	A. baumannii	blaOXA-100 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	NA	
33	A. baumannii	blaOXA-104 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17850339	
34	A. baumannii	blaOXA-106 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17850339	
35	A. baumannii	blaOXA-107 (blaOXA-51/oxaAb family)	core	Carbapenems	nwt R	17850339	Narrow spectrum carbapenem resistance - confers resistance to carbapenems if the expression enhanced by the insertion of an insertion sequence (often ISAb1, ISAb125 etc)
36	A. baumannii	blaOXA-108 (blaOXA-51/oxaAb family)	core	Carbapenems	wt S	17850339	

- Currently 396 variants in the BLDB
- Carbapenem-resistance associated with IS element upstream and certain variants
 - Probable gene duplication
- How do we interpret this with short read data?

Versatile resistance determinant IS*Aba1*

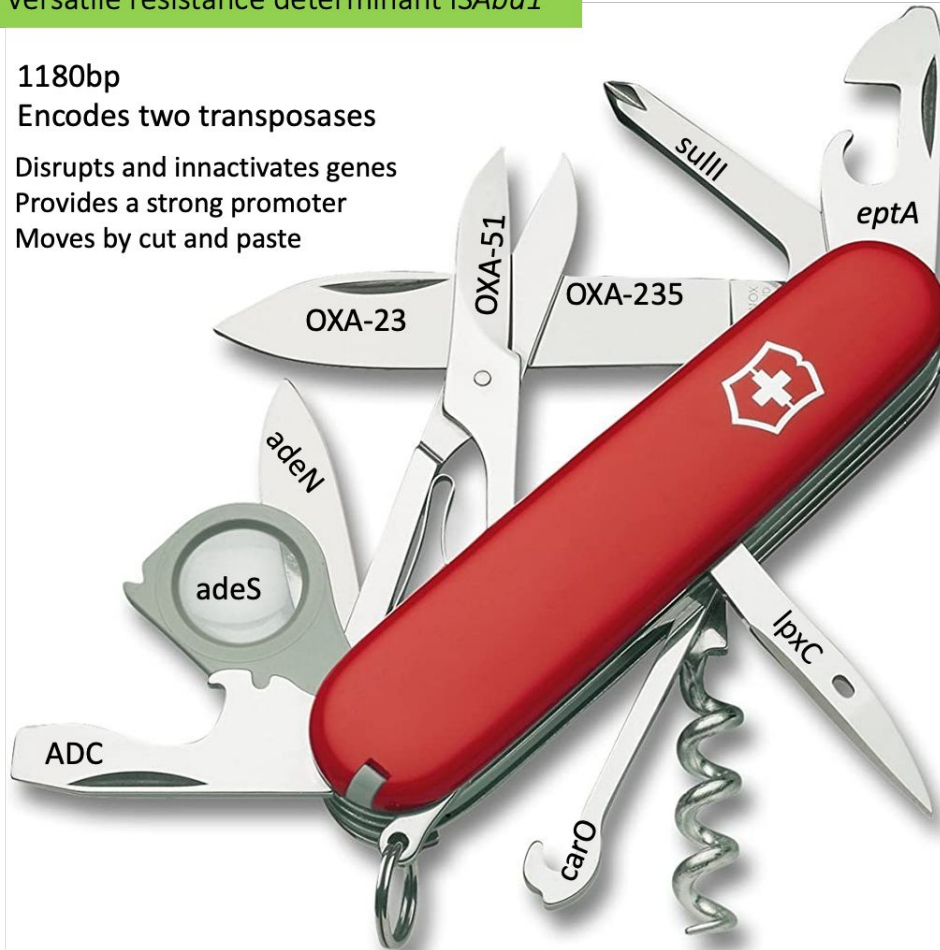
1180bp

Encodes two transposases

Disrupts and inactivates genes

Provides a strong promoter

Moves by cut and paste



Target	Antibiotic(s)	Reference
<i>bla</i> _{ADC}	cephalosporin	JAC. 2003. Vol 52. pp 629-635
<i>bla</i> _{OXA-51}	carbapenems	FEMS. 2006. Vol 258. pp 72-77
<i>bla</i> _{OXA-23}	carbapenems	Plasmid. 2003. Vol49. pp 169-178
<i>bla</i> _{OXA-235}	carbapenems	AAC. 2013. Vol 57. pp 2121-26
<i>adeS</i> (ABC)	MD efflux	PLOS One. 2012. Vol 7. e49534
<i>adeN</i> (IJK)	MD efflux	Virulence. 2017. Vol 8. pp1088-90
<i>eptA</i>	colistin	JAC. 2019. vol 74. pp2544-2550
<i>lpxC</i>	colistin	Frontiers. 2017. Vol 7. Article 45
<i>sulII</i>	sulphonamide	FEMS. 2005. Vol 253. pp 425-9

<https://www-is.biotoul.fr/>

Next steps

We have 406 rules;

- Rules for intrinsic OXA-51-like, ADC, *ant(3'')-II*, *abaF*, *craA*
- Rules for *gyrA* and *parC* mutations
- Rules for acquired OXA-23-like, OXA-24/40-like, OXA-58-like, OXA-143-like, OXA-235-like, GIM-variants, IMP-variants, NDM-variants, TMB, VIM-variants, KPC-variants

Next steps

We have 406 rules;

- Rules for intrinsic OXA-51-like, ADC, *ant(3'')-II*, *abaF*, *craA*
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To-do

- GES, PER, SHV, TEM, VEB, CTX-M, narrow-spectrum and ESBL-OXAs
- RND efflux pumps
- Aminoglycoside modifying enzymes
- Colistin
- Cefidericol
- IS elements (particularly *ISAbal*) disrupting regulators, ORFs....

Legionella

Ghislaine Descours, Charlotte Michel, Christophe Ginevra, Stefano De Giorgi, Nancy Flountzi, Sophie Jarraud

Update from *Legionella* subgroup

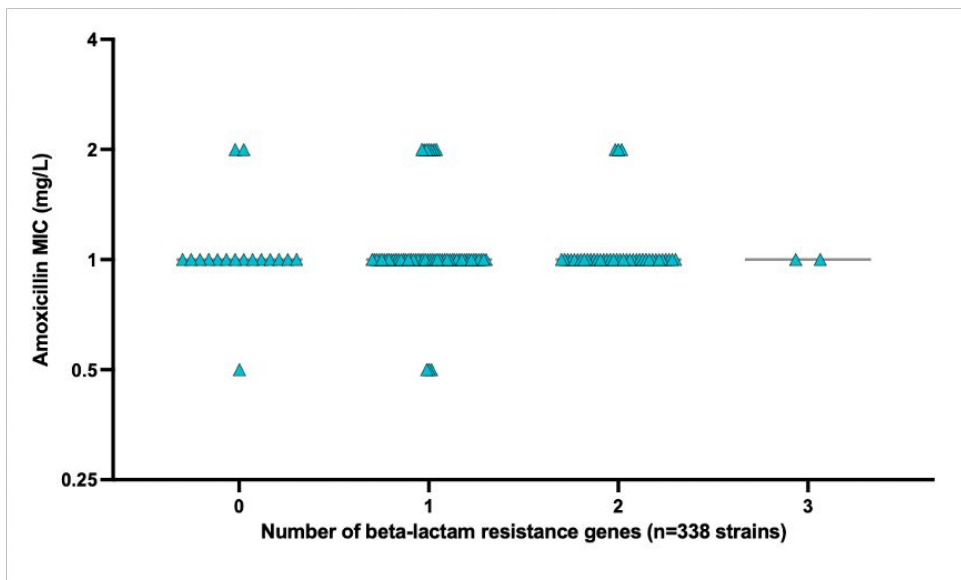
Ghislaine Descours, Charlotte Michel

- Only 2 macrolide-resistant strains isolated so far
- Main issues
 - AST is not usually performed for *Legionella*
 - Tentative EUCAST ECOFF values, 2021: « Sufficient data to establish ECOFFs are not currently available but the published MIC ... give an indication of MICs for wild type isolates by the two methods presented in this document » (gradient test on BCYE, BMD)
(https://www.eucast.org/fileadmin/src/media/PDFs/EUCAST_files/Guidance_documents/Legionella_guidance_note_-_20210528.pdf)
 - No clinical breakpoints
- Rules submitted in Feb 2026
 - 63 rules defined for **macrolides**, **fluoroquinolones**, rifampicin, tetracyclines, beta-lactams
 - Mainly **core** genome (n=60): 23S rRNA, *rplD/V*, *gyrA/B*, *parC*
 - Mainly from **experimentally evolved** mutants (+ reconstructed mutants: strong evidence)

- **Missing?**

- LoxA: *L. pneumophila* Philadelphia, OXA family (Avison 2002)
- FEZ-1: *L. gormanii* ATCC 33297 (Boschi 2000)

But no impact of presence/absence of beta-lactam determinants in clinical setting and on MICs

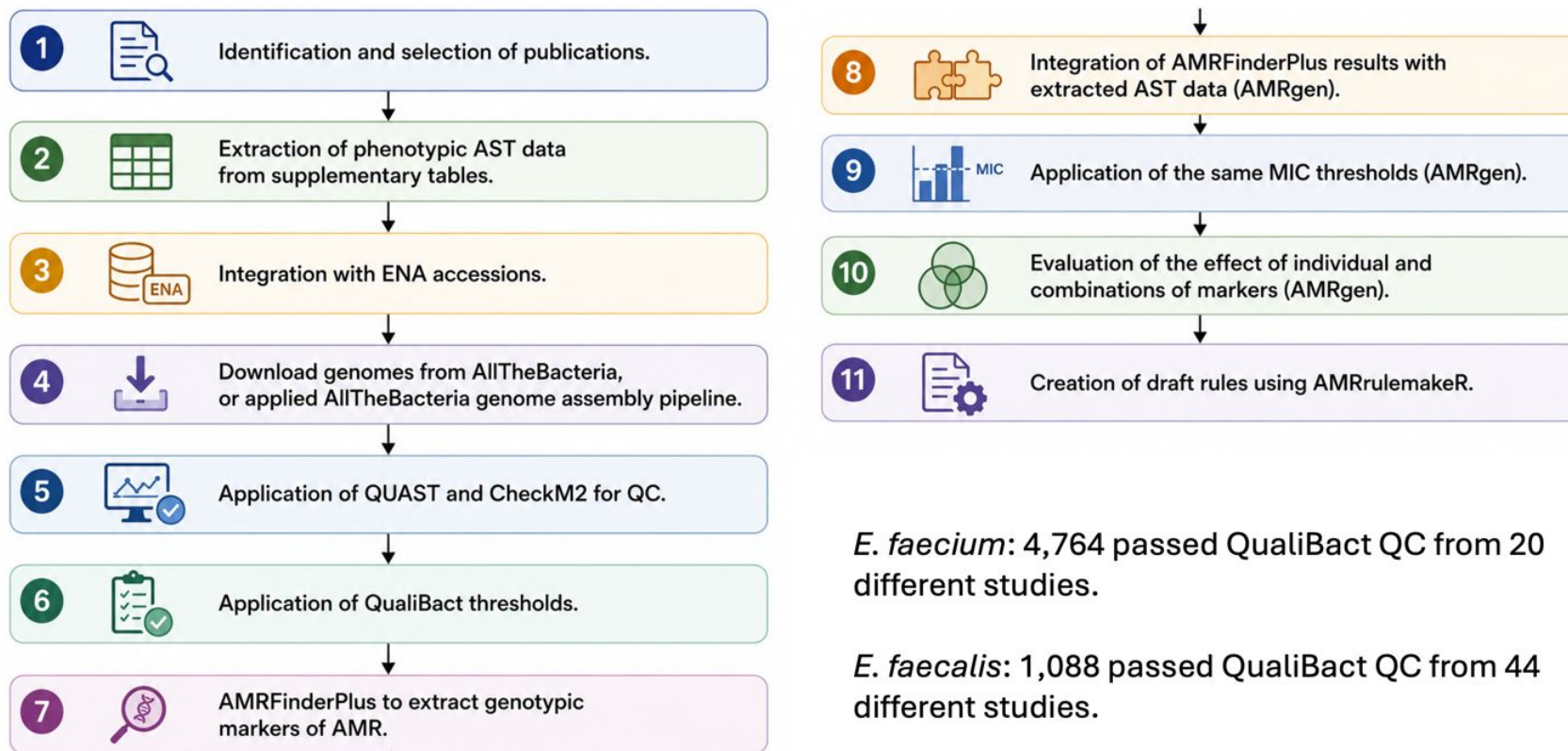


Ongoing work, French Ref Lab)

Enterococcus

Francesc Coll, Thomas Demuyser, Ana R. Freitas, Guido Werner, Precious Osadebamwen, Theo Gouliouris, Fiona Walsh, Valeria Bortolaia, Basil Britto Xavier, Helena Seth-Smith

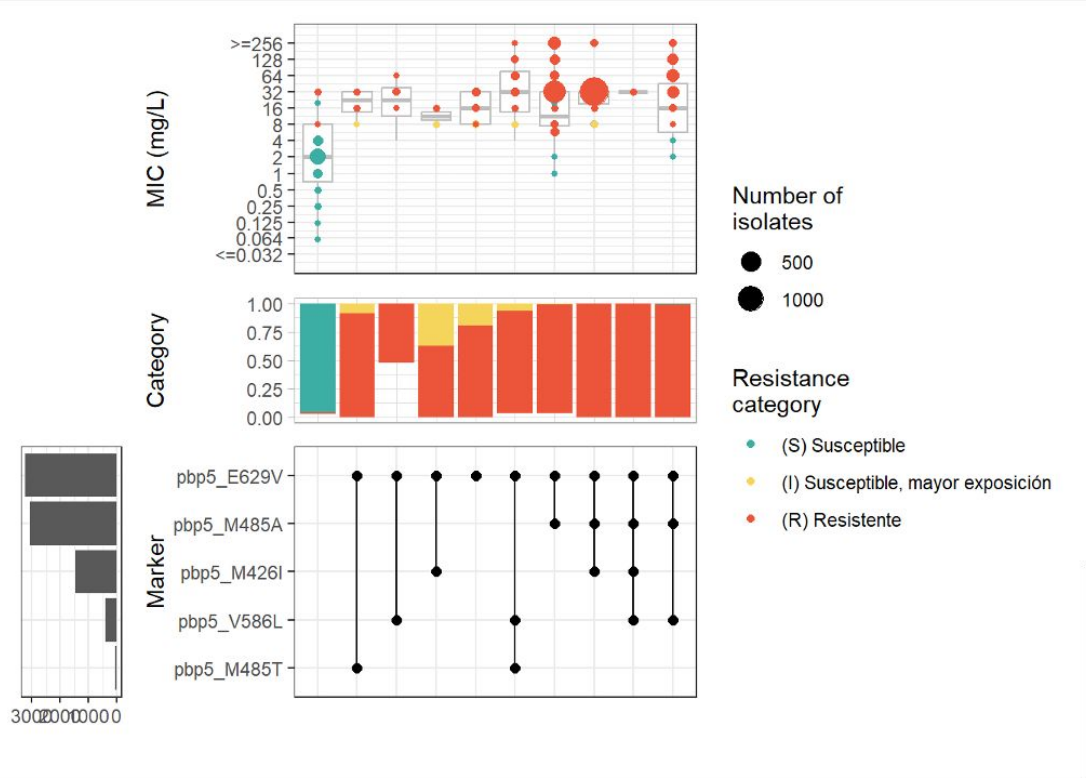
Defining rules for acquired resistance in enterococci: datasets & analytical workflow



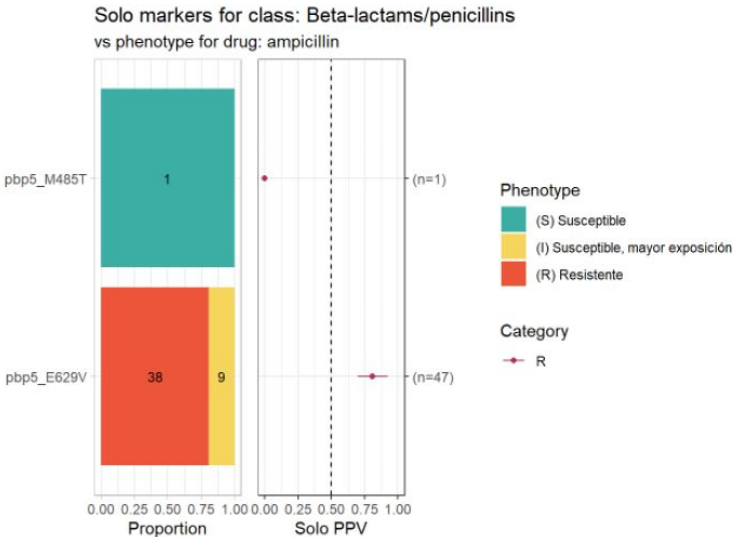
E. faecium: 4,764 passed QualiBact QC from 20 different studies.

E. faecalis: 1,088 passed QualiBact QC from 44 different studies.

Defining rules for acquired ampicillin resistance in *E. faecium*



With the exception of Pbp5 E629V, all other mutations occurred in combination, and increased MIC, except for Pbp5 M426I.



Effect on MIC

Base: 2.05

pbp5_E629V: 7.19

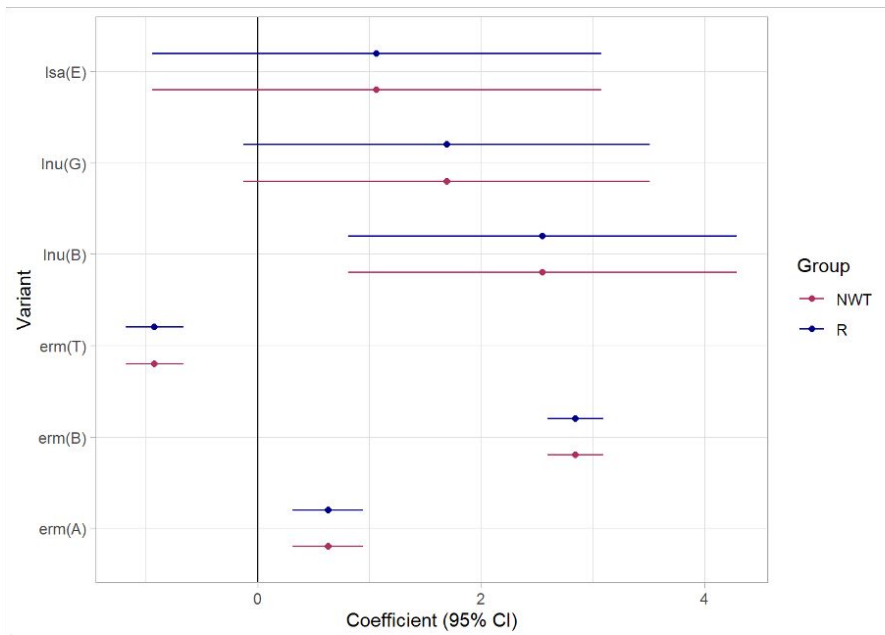
pbp5_M485A: 2.46

Pbp5_M485T: 1.77

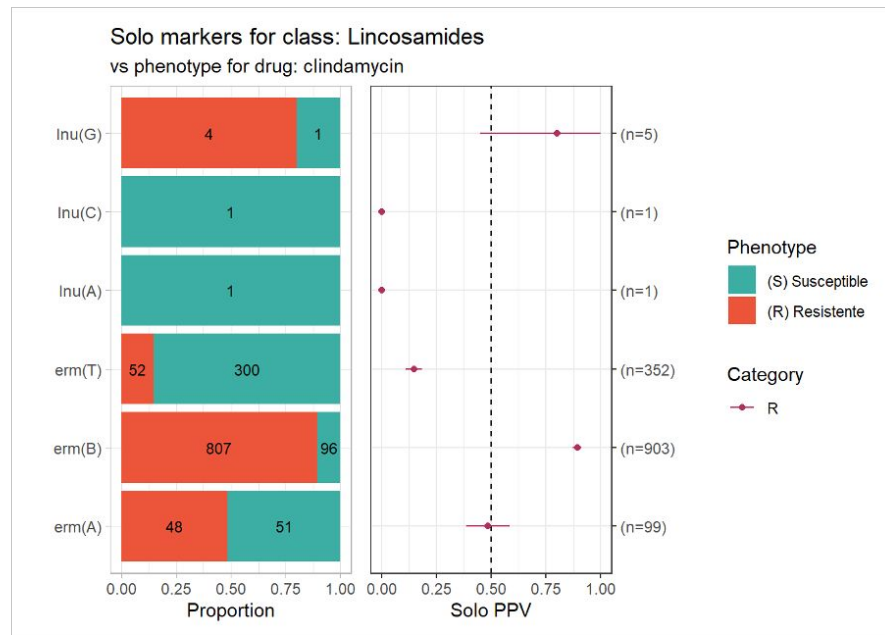
Pbp5_V586L: 1.47

Pbp5_M426I: 0.89

Defining rules for acquired ampicillin resistance in *E. faecium*



Genotype-phenotype data rules out the effect of certain markers (*erm(T)*) on clindamycin resistance. *erm(A)* seems to confer lower level of resistance than *erm(B)*, also for erythromycin.



Effect on MIC

Base: 0.99

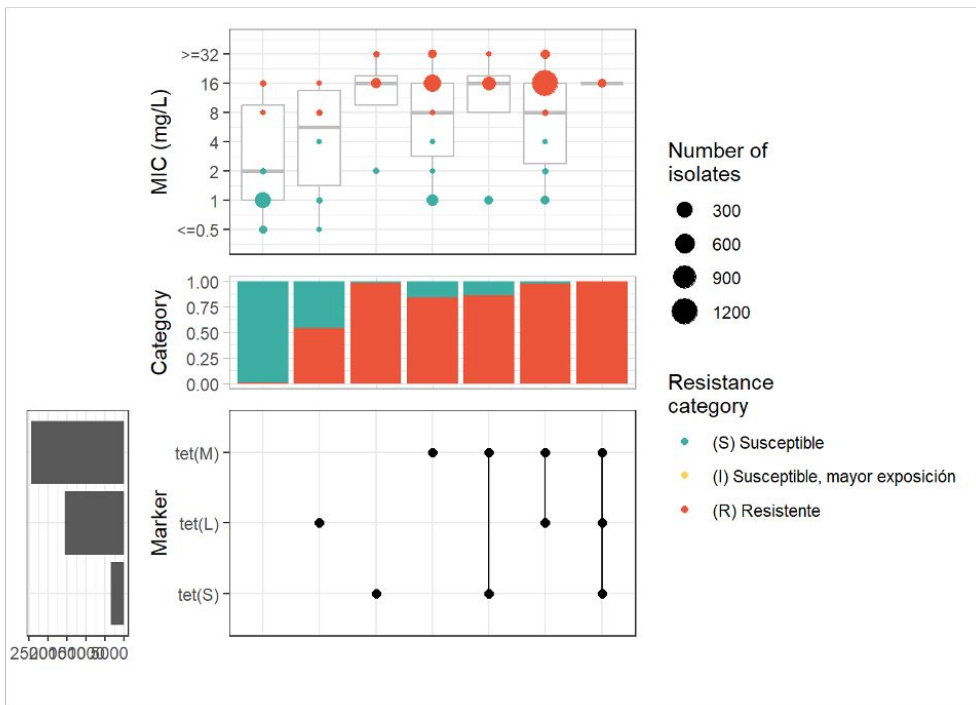
erm(B): 2.54

lnu(B): 1.67

erm(A): 1.36

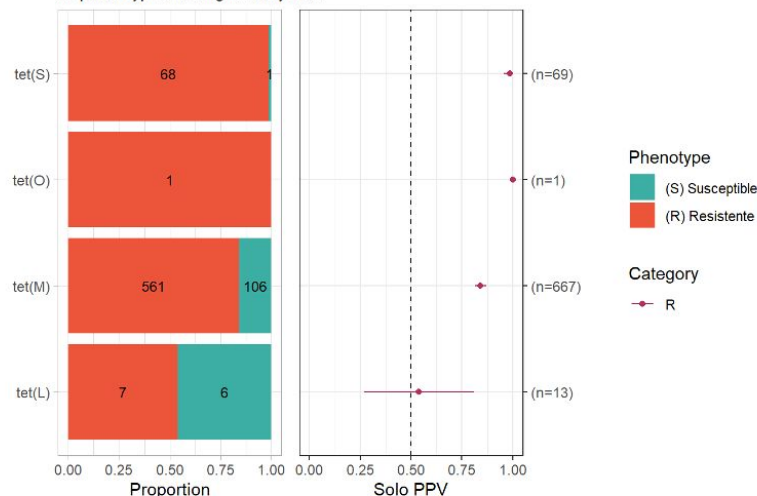
erm(T): 0.60

Defining rules for acquired tetracycline resistance in *E. faecium*



Most *tet* genes occur in combination, but also in isolation, so Solo PPV can be calculated.

Solo markers for class: Tetracyclines
vs phenotype for drug: tetracycline



Effect on MIC

Base: 1.37

Tet(L): 1.71

Tet(M): 6.50

Tet(S): 1.93

Summary of results so far, and next steps

- Rules explaining **intrinsic resistance** to aminoglycosides (determined by *aac(6')-I* and *efmM*), macrolides (*msr(C)*) and cephalosporins (*pbp5*) in *E. faecium*;
- Rules explaining **intrinsic resistance** to cephalosporins (*pbp4*), clindamycin (*Isa(A)*) and quinupristin-dalfopristin (*Isa(A)*) in *E. faecalis*.
- To define rules for **acquired resistance**, we compiled a dataset of 4,764 *E. faecium* and 1,305 *E. faecalis* genomes with available AST data.
- Using AMRgen, we defined over **100 draft rules** for acquired resistance in *E. faecium*, covering 82 unique genetic markers across 14 different antibiotics
- Using AMRgen, we defined over **35 draft rules** for acquired resistance in *E. faecalis*, including 25 unique genetic markers and 12 antibiotics.
- The association between individual markers and phenotypic resistance could not always be determined due to frequent co-occurrence of multiple AMR markers, or was statistically weak due to the low prevalence of markers or phenotypic resistance.
- We will next integrate these results with epidemiological and molecular evidence from the scientific literature to established the evidence strength supporting these draft rules.

Achromobacter xylosoxidans

Charlotte Michel, Lucie Amoureux, Miglė Gabrielaitė, Lisa Pählman

Burkholderia pseudomallei

Database initiation: We have generated first version of rules for antibiotic non-susceptibility associated with key frontline antibiotics recommended for treating melioidosis. Many thanks to @Thawanrat Khuanton for taking on much of the heavy lifting on this.

Database application: We plan to use this database to map the distribution of these variants across environmental samples and clinical isolates. This will allow us to monitor their emergence and decline over time. Also, we plan to explore signals overlapping with ongoing GWAS analyses, including associations with host mortality.

Claire Chewapreecha, Derek Sarovich, Jessica Webb, Chalita Chomkatekaew, Thawanrat Khuanton, Arin Wongprommoon

Neisseria gonorrhoeae

Leonor Sanchez Buso, Yonatan Grad, Sheeba Manoharan-Basil, Martin McHugh, Tatum Mortimer, Anna Roditscheff, Faina Wehrli, Adam Witney, Raffael Frei, Daniel Golparian, Magnus Unemo

- Revising first set of rules with combinations of mutations, and adapting them to results from AMRrulemakeR.
- Starting to plan a subgroup manuscript.

Agenda

1. Updates from subgroups
- 2. Rules with no experimental/statistical evidence**
3. Code updates
4. Manuscript planning

Rules with no experimental/statistical evidence

ruleID	txid	organism	gene	nodeID	mutation	variation type	gene context	drug	drug class	phenotype	clinical cate	breakpoint	breakpoint st	breakpoint c	ecoff	ecoff standar
STA1001	1280	s_Staphyloc	parC	parC	p.Ala116Glu	Protein varia	core	Ciprofloxacin	-	wildtype	I	disk zone >=	EUCAST 2025	-	disk zone >=	EUCAST 2025
STA1002	1280	s_Staphyloc	parC	parC	p.Glu84Gly	Protein varia	core	Ciprofloxacin	-	wildtype	I	MIC <= 2 mg/	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1003	1280	s_Staphyloc	parE	parE	p.Pro585Ser	Protein varia	core	Ciprofloxacin	-	wildtype	I	MIC <= 2 mg/	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1004	1280	s_Staphyloc	parC	parC	p.Ser80Tyr	Protein varia	core	Ciprofloxacin	-	wildtype	I	MIC <= 2 mg/	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1005	1280	s_Staphyloc	parC	parC	p.Glu84Lys	Protein varia	core	Ciprofloxacin	-	wildtype	I	MIC <= 2 mg/	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1006	1280	s_Staphyloc	parC	parC	p.Ser80Phe	Protein varia	core	Ciprofloxacin	-	wildtype	I	MIC <= 2 mg/	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1007	1280	s_Staphyloc	parE	parE	p.Pro451Ser	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	MIC <= 2 mg/	EUCAST 2025
STA1008	1280	s_Staphyloc	gyrA	gyrA	p.Ser84Leu	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1009	1280	s_Staphyloc	gyrA	gyrA	p.Gly106Asp	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	-	-
STA1010	1280	s_Staphyloc	gyrA	gyrA	p.Ser84Val	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1011	1280	s_Staphyloc	gyrA	gyrA	p.Ser85Pro	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	-	-
STA1012	1280	s_Staphyloc	parC	parC	p.Glu84Val	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	-	-
STA1013	1280	s_Staphyloc	parE	parE	p.Asp432Asn	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	-	-
STA1014	1280	s_Staphyloc	gyrA	gyrA	p.Glu88Lys	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1015	1280	s_Staphyloc	gyrA	gyrA	p.Ser84Ala	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1016	1280	s_Staphyloc	parE	parE	p.Asp432Val	Protein varia	core	Ciprofloxacin	-	wildtype	I	-	-	-	-	-
STA1017	1280	s_Staphyloc	parE	parE	p.Asn470Ser	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1018	1280	s_Staphyloc	parE	parE	p.Asp432His	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1019	1280	s_Staphyloc	gyrA	gyrA	p.Glu88Gly	Protein varia	core	Ciprofloxacin	-	-	-	-	-	-	-	-
STA1020	1280	s_Staphyloc	STA1004 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC <= 2 mg/	EUCAST 2025
STA1021	1280	s_Staphyloc	STA1005 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1022	1280	s_Staphyloc	STA1006 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1023	1280	s_Staphyloc	STA1006 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1024	1280	s_Staphyloc	STA1004 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1025	1280	s_Staphyloc	STA1005 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1026	1280	s_Staphyloc	STA1003 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1027	1280	s_Staphyloc	STA1006 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025
STA1028	1280	s_Staphyloc	STA1005 & ST	-	-	Combination	-	Ciprofloxacin	-	nonwildtype	R	MIC > 2 mg/L	EUCAST 2025	-	MIC > 2 mg/L	EUCAST 2025

Rules with no experimental/statistical evidence

Encode a rule and grade it **very low**

ruleID	txid	organism	gene	nodeID	mutation	Variation type	Gene context	drug	phenotype	Clinical category	Evidence grade
STA1008	1280	s__Staphylococcus aureus	gyrA	gyrA	p.Gly106Asp	Protein variant detected	core	Ciprofloxacin	wildtype	I	very low

Do not encode a rule

ruleID	txid	organism	gene	nodeID	mutation	Variation type	Gene context	drug	phenotype	Clinical category	Evidence grade
STA1008	1280	s__Staphylococcus aureus	gyrA	gyrA	p.Gly106Asp	Protein variant detected	core	Ciprofloxacin	-	-	very low

Note: marker has only ever been found in combination with others, so we lack evidence of its specific effect

Agenda

1. Updates from subgroups
2. Rules with no experimental/statistical evidence
- 3. Code updates**
4. Planning
 - Manuscript plans, ESCMID 2027

Code updates

AMRrulemakerR

- Minor updates, now has Vignettes (tutorial)

Public geno/pheno data

Update to spec v1.1 (with CARD ARO & mechanism)

Agenda

1. Updates from subgroups
2. Rules with no experimental/statistical evidence
3. Code updates
- 4. Manuscript planning**

2026 Planning - Manuscripts

1. **AMRgen R package** - *under review at npj AMR*
2. **Call to action for submitting AST data** - *to be submitted July*
3. **AMRrules and core gene rules** - *to be distributed July*
 - Concept
 - Why interpreting core gene rules is important but challenging
 - Approach to defining rules
 - Summary of rules - comparison across pathogens, gaps in knowledge
4. **Quantitative approach and acquired gene rules (Q4)**

A call for sharing antimicrobial susceptibility phenotype data alongside pathogen genome sequence data

Kathryn Elizabeth Holt^{1,2,3,4}, Matthijs Simeon Berends^{3,4,5,6}, Natacha Couto^{1,3,7}, Donald Brody Duncan⁸, Zoe Anne Dyson^{1,3,9}, Ebenezer Foster-Nyarko^{1,3}, Jane Hawkey^{2,3,4,10,11}, John Andrew Lees¹², Finlay Maguire^{3,13,14}, Arjun Balmiki Prasad¹⁵, **ESGEM-AMR Working Group**

ESGEM-AMR members invited to join as group co-authors

A call for sharing antimicrobial susceptibility phenotype data alongside pathogen genome sequence data

Perspective

- Depositing WGS in INSDC databases (e.g. NCBI, EBI) is a requirement of many funders, publishers, and surveillance programs
- Here we call on the global scientific community to normalise the practice of depositing accompanying AST data when submitting WGS data to public archives

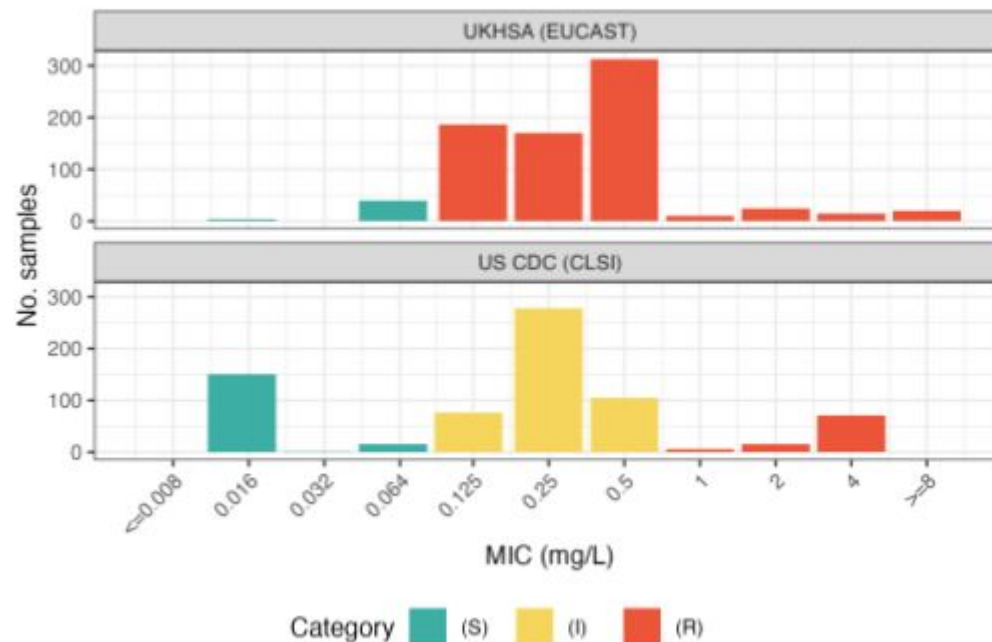
Discuss key practical considerations for sharing AST data, including data standards and formats, and highlight new tools to support preparing AST data for sharing.

Why share?
What to share?
Where to share?
How to share?

NCBI	EMBL-EBI	GLASS, EARS-Net
biosample_accession	accession	
antibiotic	antibioticName	antibiotic
resistance_phenotype	resistancePhenotype	SIR ResultMICSIR ResultZoneSIR ResultEtestSIR
measurement_sign	measurementSign	ResultMICSign ResultZoneSign ResultEtestSign
measurement	measurement	ResultMICValue ResultZoneValue ResultEtestValue
measurement_units	measurementUnits	*
laboratory_typing_method	laboratoryTypingMethod	-
laboratory_typing_platform	platform	
vendor	-	
laboratory_typing_method_version_or_reagent	-	
testing_standard	astStandard	ReferenceGuidelinesSIR
-	breakpointVersion	-
-	-	DiskLoad

Why share?
What to share?
Where to share?
How to share?

	Interpreted using local standard	Re-interpreted with consistent standard	
		EUCAST (R >0.06 mg/L)	CLSI (R >=1 mg/L)
UKHSA	94.4% (EUCAST)	94.4%	8.8%
US CDC	12.9% (CLSI)	76.8%	12.9%



How to share?

AMRgen R package, or AMRgen Converter web app
...to convert AST data to EBI or NCBI format to submit



Upload antibiogram file

Browse...

No file selected

Input format

Sensititre

Instructions

Imported data

Summary

Exports

NCBI

EBI table

EBI JSON

Export a TSV file suitable for submission to NCBI. See [NCBI BioSample Antibigrams](#) for details.

Import data to choose a phenotype column for export.

 **Download NCBI TSV**

Guideline

EUCAST

Vendor (optional)

<http://apps.amrverse.org/amrgenconverter/>

ESGEM-AMR members invited to join as group co-authors

- Go to https://bit.ly/ast_sharing (or use the QR code)
- Review the paper in Google Docs
- If you want to join as a co-author:
 1. Provide any edits/comments in the doc
 2. Complete your details in this form
- Deadline: July 10, 2026



Global diversity and antimicrobial resistance of typhoid fever pathogens: Insights from a meta-analysis of 13,000 *Salmonella* Typhi genomes

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AST data sharing paper



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Questions? / Any other business?

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